Point-by-Point Response to the Reviewers

We thank the editor and reviewers for their careful evaluation of our manuscript. We have revised the manuscript and Supporting Information in response to the comments, as detailed below.

Reviewer 1

Recommendation: Publish as is; no revisions needed.

Comments: Using a frequency-modulation atomic-force-microscopy setup, the authors simultaneous measure slip length and surface topography with nanoscale lateral resolution on different substrate types. The extracted slip lengths on flat substrates are found to be almost zero, in agreement with scaling analysis combined with molecular dynamics simulation predictions. Large apparent slip lengths are explained in terms of surface bubbles. The paper is well written and settles some long-standing controversies in the field. I therefore recommend publication as is and only have two minor comments.

We sincerely thank the reviewer for the positive evaluation of our work. We respond to the two comments below.

1. The prefactor C of equation 2 has units of length and is given by $C = 0.41$ nm.

Thank you for pointing this out. We revised the manuscript to clarify that the prefactor C in Eq. 2 has dimensions of length. We have added units to all occurrences of the scaling prefactor C in the manuscript and Supporting Information, including the figure captions and comparison table.

Changes:

- Clarified in the main text that C is a prefactor with dimensions of length. (lines 41–43)
- Updated the scaling-law prefactors to include nanometer units. (lines 48–52)
- Updated the corresponding units in the Fig. 1 caption. (line 54)

2. The scaling law in equation 2 was derived for non-polar surfaces that interact with water only via dispersion or Lennard-Jones interactions, that means that hydrophilic surfaces were created by large dispersion attraction. In the recent publication "Subnanometer Interfacial Hydrodynamics: Spatially Resolved Viscosity and Surface Friction" Carlson et al., Nano Lett. 2025, 25, 15605–15612 MD simulations were used to model polar surfaces where small contact angles are created by polar surface groups, which is more realistic. The fit of the friction coefficient is for these systems better represented by an exponential, not by a power law. For the non-polar surfaces the authors are mostly interested this does not make a difference, but for future studies this could be relevant.

Thank you very much for your helpful comment. Our method is also capable of measuring polar surfaces, and we are in fact planning to study such surfaces in the near future. In that context, the paper you pointed out will be very useful. Thank you again for bringing it to our attention.

Reviewer 2

Recommendation: Publish after major revisions noted.

Comments: This is an interesting article that deals with experimental measurements of the water slip length on various surfaces, coming to the conclusion that water need not slip on hydrophobic surfaces. The measurements are made using frequency-modulation atomic force microscopy (AFM), with claims of a 159-fold improvement in the sensitivity of measurement. The authors found a large slip length of 43.2 nm on graphite in deionized water. Overall, the manuscript can be of interest to a broad cross-section of researchers in the field of nanoscience and nanotechnology. Thus, the manuscript may eventually be suitable for publication in Nano Letters. However, at this juncture, several critical points require careful consideration and revision in the manuscript, before the manuscript can be considered for publication. I outline these points below:

We sincerely thank the reviewer for the careful reading of our manuscript and for the constructive comments. We respond to each point below.

> **1.** Firstly, the title of the paper is currently not wholly accurate. The word “Does” should be replaced with “Need”, i.e., the title should be “Water Need Not Slip on Hydrophobic Surfaces: Insights from Slip Length Mapping”. This is because water does slip on graphite, so the authors cannot generally state that water does not slip on hydrophobic surfaces. Instead, the claim should be that water need not slip on hydrophobic surfaces.

Thank you for this constructive suggestion. We agree that the title needs clarification. We have revised the title to “Hydrophobicity Does Not Determine Water Slip Length: Insights from Slip Length Mapping” to clarify that wettability alone does not determine slip length on flat surfaces, while acknowledging that multiple factors influence interfacial friction.

Changes:

- Revised the manuscript title from “Water Does Not Slip on Hydrophobic Surfaces: Insights from Slip Length Mapping” to “Hydrophobicity Does Not Determine Water Slip Length: Insights from Slip Length Mapping”.

> **2.** The authors need to contextualize their results vis-a-vis those of Secchi et al. (`https://www.nature.com/articles/nature19315`), who found massive radius-dependent water slippage in carbon nanotubes (CNTs), with slip lengths on the order of hundreds of nanometers. The authors should explain why there is massive slippage on CNTs but not on the surfaces they examined.

Thank you for this important suggestion. We now clarify that the massive, radius-dependent slippage inside CNTs does not contradict our results on nanoscopically flat planar surfaces. As Secchi et al. demonstrated, the slip length can reach several hundred nanometers due to a curvature-enhanced effect, whereas it is much smaller on a non-curved plane, on the order of several tens of nanometers. Our results on planar HOPG are quantitatively consistent with the slip length reported by Secchi et al. for planar graphite. We therefore distinguish the planar graphite slip regime observed here from the confinement-induced CNT transport regime.

Changes:

- Added a new paragraph comparing the present planar HOPG result with Secchi et al.’s CNT slippage results. (lines 217–222)

> **3.** The derivation of eq. 3 and 4 must be provided along with the paper and the symbol hdot must be explained in the main text, including how it is measured experimentally.

Thank you for pointing this out. We have revised the main text to clarify the origin of Eqs. 3 and 4 and to define $\dot{h}$. Equation 3 is the lubrication-theory expression for the hydrodynamic force between a sphere and a plane with slip boundary conditions, and we now cite the original theoretical work by Vinogradova for this result. Because the full derivation is lengthy for a Letter journal, we refer readers to the original theoretical work. For the same reason, we have provided the derivation of Eq. 4 in the Supporting Information. To help readers follow the main argument, we have added a concise explanation in the main text stating that Eq. 4 follows from the relation $\gamma_{\mathrm{total}} = \gamma_{\mathrm{tip}} + \gamma_{\mathrm{bulk}}$, $Q = k/(\omega\gamma_{\mathrm{total}})$, and the proportionality $A \propto Q$ under constant drive power. A more detailed derivation is beyond the scope of the present Letter, and we plan to present it separately in a future paper. We also now define $\dot{h}$ as the time derivative of the measured probe-substrate distance $h(t)$, calculated numerically from the measured $h(t)$ data.

Changes:

- Clarified the origin of Eq. 3 and defined $\dot{h}$ in the main text. (lines 71–78)
- Added a concise derivation route for Eq. 4 and pointed to the Supporting Information for details. (lines 92–94)

4. The authors mention that the "Slip length bs is then obtained by fitting Eq. 5 to a damping-coefficient-tip-sample distance curve with bs as the fitting parameter." However, the functional form of f^* in Eq. 5 is not provided, making it unclear as to how bs can be extracted. This omission must be clarified and the form of f^* must be derived and mentioned in the paper.

Thank you for pointing out this omission. We have revised the main text to explicitly provide the functional form of the correction factor f^*, which was used in Eqs. 3 and 5 in the original manuscript. Specifically, $f^*(h, b_t, b_s)$ is now given as Eq. 4, and the auxiliary quantities A, B, and C are defined in Eq. 5 of the revised manuscript.

Changes:

- Added the explicit functional form of $f^*(h, b_t, b_s)$ and defined A, B, and C. (lines 78–80)
- Updated the notation from $f^*(b_t, b_s)$ to $f^*(h, b_t, b_s)$ in Eqs. 3, 7, S7, and S8.

5. For the fit in Figure 2a, please provide the goodness of fit (R^2) or any other metric, such as the mean absolute error, to quantify how good the fit is. Also, the authors must explain as to how error bars were obtained on the reported slip lengths. Are these confidence intervals from the fit, or standard deviations/errors from repeated measurements?

Thank you for this helpful suggestion. Because the R^2 value is not necessarily suitable for evaluating nonlinear fitting, we have added the mean absolute error for the representative fit in Fig. 2a. The fitted curve gives a mean absolute error of $0.014\,\mu\mathrm{N\,s\,m^{-1}}$. The small absolute residual shows that the fitted curve quantitatively captures the damping reduction relative to the no-slip prediction. In addition, we clarified that the error bars in Fig. 4 are standard deviations of the slip lengths obtained from the 128×128 pixels in each slip-length map, rather than confidence intervals from a single fit or standard errors.

Changes:

- Added the mean absolute error for the fit in the caption of Fig. 2. (line 107)
- Clarified in the Fig. 4 caption that the error bars are standard deviations calculated from the pixel-wise slip-length values in each map.

6. The authors state that "We calibrated the slip length of the probe tip by measuring a symmetric system ($b_t = b_s$) using an atmospheric-plasma-treated diamond-like carbon (DLC) substrate, yielding a slip length of $b_t = b_s = 0.0 \pm 1.2$ nm."

7. The authors should show the fit plot for this case, similar to what they did in Figure 2a. Plots should also be shown for the fit used to extract the slip length of 345.3 nm above the nanobubble, as mentioned towards the end of page 8.

Thank you for the suggestions. We have added representative fitting plots for the atmospheric-plasma-treated DLC substrate and nanobubble, as shown in Fig. R1(a) and (b), respectively. The plots show the measured damping coefficient, the curve for the case of no slip, and the fitted curve. The mean absolute errors of these fits were $0.016\,\mu\mathrm{N\,s\,m^{-1}}$ and $0.014\,\mu\mathrm{N\,s\,m^{-1}}$, respectively.

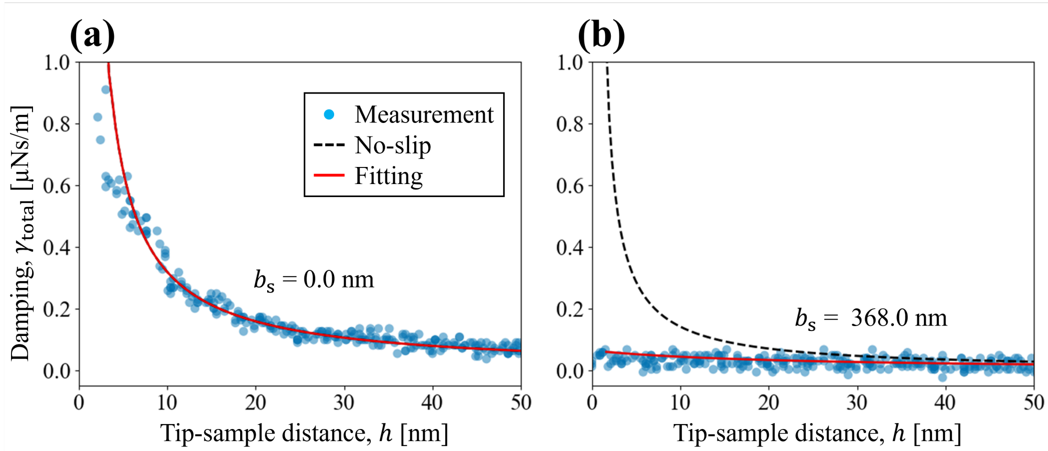

Figure R1: Representative damping-coefficient fitting curves for (a) an atmospheric-plasma-treated DLC substrate and (b) a location above a nanobubble.

Changes:

- Added the representative fitting plots for DLC and nanobubble (Fig. R1) as Fig. S2(a, b) in the Supporting Information.
- Added a main-text reference to the new DLC fitting plot. (line 143)
- Added a main-text reference to the new nanobubble fitting plot. (lines 160–161)

> **8.** Why are negative slip lengths obtained on the flat silica and FOPA regions? What is the physical meaning of a negative slip length? Isn't the lowest value of the slip length 0?

Thank you for raising this point. A negative slip length formally means that the extrapolated zero-velocity plane lies on the liquid side of the geometrical solid-liquid interface. In the present measurements, we do not interpret the small negative values on silica and FOPA as physically meaningful negative slip. Rather, they likely reflect small apparent offsets caused by the propagation of systematic uncertainties in the probe radius, spring constant, and distance origin into the fitted b_s. Because all surfaces other than HOPG consistently yielded slip lengths close to zero, these offsets are sufficiently small compared with the slip length observed on the slippery surface.

Changes:

- Added an explanation of the formal meaning of negative slip length and clarified that the measured small negative values are apparent offsets caused by error propagation. (lines 173–179)

> **9.** Why is the water slip length 0.8 nm on Teflon but 43.2 on HOPG, despite Teflon expected to have a higher contact angle of water than HOPG?

Thank you for raising this important point. This is precisely one of the key claims of our paper. More specifically, the difference in slip length between Teflon and HOPG cannot be explained by the contact angle. Slip length is governed by interfacial friction, which depends strongly on the atomic-scale structure and chemical uniformity of the solid-water interface. Although Teflon is more hydrophobic, it does not provide the atomically smooth and chemically homogeneous crystalline interface characteristic of graphite. We therefore attribute the large slip length on HOPG not to wettability, but to the low-friction graphite-water interface. Several previous MD simulations have suggested the importance of atomic-scale geometrical and chemical homogeneity at the interface (Ref. 19) and the independence of wettability on solid surfaces with contact angles below 120° (Ref. 16). To the best of our knowledge, however, our study is the first to experimentally verify these predictions. This point is discussed in lines 196–216 of the revised manuscript.

10. Figure 4 presents a systematic comparison of slip lengths on nanoscopically flat substrates. How is it proved that these surfaces are nanoscopically flat? The authors need to provide some proof using AFM measurements.

We apologize for the confusion. AFM topography maps for the nanoscopically flat substrates are provided in the Supporting Information as Fig. S3, together with the corresponding slip-length maps. We have revised the Fig. 4 caption to explicitly direct readers to these topography data and the measured nanoscale roughness values. The measured R_a values were 0.27 nm, 0.29 nm, 0.64 nm, and 0.35 nm for mica, FDTS, Teflon, and HOPG, respectively, confirming that these substrates are nanoscopically flat.

Changes:

- Revised the Fig. 4 caption to explicitly refer to the Supporting Information topography maps and measured nanoscale roughness values.

11. For eq. 2, the units of b must be specified for completeness. The authors also must provide a table of the calculated b (from eq. 2) and the measured b from their experiments for all the surfaces examined in this work.

Thank you for pointing this out. We have revised the manuscript to clarify that both b and the prefactor C in Eq. 2 have dimensions of length. We have also expanded Table S2 of the Supporting Information to compare, for every substrate investigated in this study, the experimentally measured slip length with the values predicted by Eq. 2 using both scaling constants.

Changes:

- Added a main-text reference to the comparison table. (lines 194–196)
- Added slip lengths predicted by both scaling constants in Supporting Information Table S2. (SI line 74)

12. How can the authors claim that a water contact angle of 120 degrees is the near-maximum achievable on a topographically smooth flat surface? Ref. 21 mentions that "This value is considered to be the lowest surface free energy of any solid, based on the hexagonal closed alignment of –CF3 groups on the surface." Thus, for surfaces with other than CF3 groups, the contact angle could be higher. Also, the presence of CF3 groups would make the surface topographically rough, as compared to say, graphite.

Thank you for the valuable comment. As noted in Ref. 21, the $-CF_3$ group has the lowest surface free energy among surface-terminating groups, and thus a surface on which $-CF_3$ groups are arranged in a hexagonal close-packed structure gives the largest water contact angles attainable on a chemically homogeneous flat surface, approximately 120°. Thus, this value is the near-maximum achievable on topographically smooth flat surfaces. To our knowledge, no chemically and geometrically homogeneous surface with an intrinsic water contact angle exceeding 130° has been reported. Also, as you noted, surfaces bearing $-CF_3$ groups are topographically rougher than graphite. However, as shown in Fig. S3, the surface roughness of Teflon is 0.64 nm, which is larger than that of graphite, 0.35 nm, but still sufficiently flat and therefore does not affect the contact angle. We also revised the caption of Fig. 4 accordingly.

Changes:

- Revised the main-text statement so that 120° is not presented as an absolute maximum. (lines 43–47)
- Revised the Fig. 4 caption to describe the gray region as an approximate upper range for chemically homogeneous flat surfaces.

13. The authors mention that "The contact angles of deionized-water droplets, measured separately on the hydrophilic and hydrophobic surfaces, were 28 degrees and 105 degrees, respectively." How quickly were these measurements done since exposure to the atmosphere can change the contact angle, particularly for hydrophilic surfaces?

Thank you for raising this point. We have revised the manuscript to clarify how the contact angles of the hydrophilic and hydrophobic regions were measured. The contact angles were measured immediately before the FM-AFM measurements. Because the patterned regions in the composite substrate were too small for contact angle measurements, we performed the measurements on unpatterned reference substrates prepared using the same surface-treatment procedures as those used for the composite substrate. We added this detail to the Supporting Information.

Changes:

- Clarified in the main text that contact angles were measured immediately before FM-AFM measurements using unpatterned reference substrates prepared by the same process as the corresponding composite-substrate regions. (lines 132–135)
- Added the corresponding procedural detail to the Supporting Information. (SI lines 59–63)

14. The authors need to substantiate the claim that "compared with hydrophilic surfaces (i.e., contact angles < 90 degree), there is no clear consensus on slip lengths for hydrophobic surfaces (i.e., contact angles > 90 degree)." They should also state what consensus there is in the field for hydrophilic surfaces, along with providing suitable references.

Thank you for pointing this out. We have revised the introduction to substantiate this statement more explicitly. The revised text now states that reported slip lengths on hydrophilic smooth surfaces are generally close to zero within experimental uncertainty, whereas values reported for hydrophobic surfaces scatter widely from near-zero to tens or even hundreds of nanometers, as summarized in Fig. 1. We also clarified that previous studies have attributed large apparent slip on hydrophobic surfaces to several different origins, including surface roughness, nanobubbles, and measurement artifacts, leaving the intrinsic slip length on hydrophobic surfaces unresolved.

Changes:

- Clarified the near-no-slip consensus for hydrophilic smooth surfaces and the broad literature scatter for hydrophobic surfaces. (lines 29–39)

15. Pristine graphite is actually hydrophilic as revealed in the last decade (`https://pubs.acs.org/doi/full/10.1021/acs.accounts.6b00447`). How do the authors then explain the relatively large water slip length seen on HOPG in their measurements? Is it due to hydrophobic molecules adsorbed on the surface, or due to nanobubbles, as they have speculated in some other cases in the paper?

Thank you for raising this important point. We revised the discussion to clarify the origin of the large slip length observed on HOPG. The HOPG surface was freshly cleaved immediately before measurement and showed moderate wettability (59.0°), close to the value reported for freshly cleaved HOPG by Kozbial et al. (64.4°). This makes hydrophobic adsorbates unlikely to be the primary origin of the large slip length. We also clarified that the localized nanobubble observed in Fig. 3(b) was analyzed separately and did not contribute to the slip length measured on the flat HOPG terrace. HOPG exhibits large slip because its atomically flat and chemically homogeneous crystalline surface forms a low-friction interface with water, as discussed in our response to question 9.

Changes:

- Clarified that the measured HOPG contact angle is close to the reported value for freshly cleaved HOPG, that hydrophobic adsorbates are unlikely to be the primary origin of the large HOPG slip length, and that the localized nanobubble was analyzed separately. (lines 204–210)

Reviewer 3

We sincerely thank the reviewer for the thoughtful comments. We respond to each point below.

1. The work seems not support the title "Water Does Not Slip on Hydrophobic Surfaces". The slip effect may be more obvious in the confined water region (such as water in the carbon nanotube), for which the Knudsen number is used to determine its impact.

2. In recent work (Nature 2022, 602 (7895), 84– 90), the quantum friction was proposed for the water-carbon interaction. The following work (Nano Lett. 2023, 23 (2), 580– 587; J. Phys. Chem. Lett. 2026, 17(10), 2792-2798) shows the thickness and defect of carbon material also have a significantly impact. All these mean the solid will have influence on the friction, which was not considered in the previous MD simulation.

3. The developed experimental method for slip length measurement in present work shows its advantage, however, the conclusion of "Water Does Not Slip on Hydrophobic Surfaces" should be reconsidered, as the friction is influenced by many factors for both solid and liquid.

We appreciate your thoughtful comments. We agree that interfacial friction depends on multiple factors including those you mention. However, our key finding is that on flat, chemically homogeneous surfaces, *wettability alone does not determine slip length.* To better reflect this, we have revised the title to "Hydrophobicity Does Not Determine Water Slip Length: Insights from Slip Length Mapping." This emphasizes that we are specifically examining the limited role of wettability on flat surfaces, while acknowledging the importance of other factors such as atomic-scale structure and confinement effects.

Changes:

- Revised the manuscript title from "Water Does Not Slip on Hydrophobic Surfaces: Insights from Slip Length Mapping" to "Hydrophobicity Does Not Determine Water Slip Length: Insights from Slip Length Mapping".

Non-scientific changes

1. Please provide full contact information for all authors in manuscript file: institution, city, state, postal code, country for each affiliation (states are required for United States addresses only). Postal codes are also required for addresses outside the U.S., for countries that have them. Each separate affiliation requires its own address information; they cannot be combined.

Thank you for pointing this out. We have provided complete contact information for all authors, including institution, city, state/region, postal code, and country for each affiliation as required.

2. An Abstract, which should summarize the reason for the work, the most significant results, and the conclusions, must be present and labeled.

Thank you for this comment. We have ensured that the abstract is clearly labeled and prominently displayed using the same section formatting as in the manuscript.

Changes:

- Labeled the abstract. (line 1)

3. Keywords are missing. Four to six keywords should be placed after the abstract.

Thank you for pointing this out. We have added keywords to the manuscript after the abstract.

Changes:

- Added the keywords. (line 15)

4. Copyright permissions are required for any graphic that is reproduced or adapted from a source that is not published by the American Chemical Society. See, for example, Figure 1. Please make sure that all graphics reproduced or adapted from another journal have the proper permission forms and credit lines in your revision.

Thank you for pointing this out. Figure 1 was prepared by adapting and extending the plot reported by Wu et al. We have obtained copyright permission from the Society of Petroleum Engineers and uploaded the permission form as "Other Files for Editors Only." We have also revised the Figure 1 caption to include the required credit line.

Changes:

- Added copyright credit line and adaptation statement to Figure 1 caption.